

Athletic Runner Injury Prediction System

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Abstract—Competitive running, with its rigorous training regimens, exposes athletes to a heightened risk of injuries. Prior approaches use simplistic models and lack data integration. Existing methods though valuable, have limitations in terms of their accuracy and precision. Hence, athlete-specific recommendations remain a challenge. The proposed method combines diverse attributes of biomechanical data, and athlete training load metrics, to build a comprehensive injury prediction model. It employs various machine learning approaches and establishes a feedback loop mechanism that enables athletes and coaches to act upon the model's predictions, adjusting training plans as needed. The system conducts a thorough comparison of various machine learning algorithms to identify and employ the most accurate algorithm. Bagging and Decision Tree consistently show better accuracy and precision, indicating their robustness in predicting athletic runner injuries. These findings underscore the importance of dataset augmentation and the selection of appropriate algorithms in enhancing model performance for injury prediction tasks.

Keywords—machine learning, running injury, machine learning, athlete training load metrics, feedback loop mechanism, injury prediction.

I. INTRODUCTION

In the realm of competitive sports, athletic runners face a unique set of challenges, with rigorous training regimens and intense competitions exposing them to a heightened risk of injuries [1]. The impact of these injuries extends beyond mere physical setbacks, often influencing an athlete's performance and, crucially, their long-term career trajectory. Traditional approaches to injury prediction within the realm of athletic running have often fallen short, relying on simplistic models or lacking the integration of comprehensive datasets. These limitations have resulted in reduced accuracy and applicability, failing to account for the dynamic nature of injury risks across training cycles, individual variations, and environmental factors. Moreover, the interpretability of predictions and the formulation of athlete-specific recommendations remain persistent challenges. This proposed system addresses the shortcomings of existing methods by combining diverse attributes, such as biomechanical data and training load metrics, to construct a robust and comprehensive injury prediction model. Leveraging various machine learning approaches, the system aims to unravel the intricate relationships within the data, enabling early and precise prediction of potential injuries. Unlike the traditional system, the proposed one incorporates a feedback loop mechanism, empowering athletes and coaches to proactively respond to the model's predictions by adapting training plans in real time.

Additionally, the system conducts an exhaustive comparison of different machine learning algorithms[2], ensuring the utilization of the most accurate and effective method for injury prediction. In striving for excellence, this Athletic Runner Injury Prediction System not only seeks to enhance the well-being of competitive runners but also aligns with broader objectives, contributing to the realization of Sustainable Development Goals by promoting healthy lives and overall well-being in the athletic community.

A. Motivation

Running-related injuries pose a significant challenge to the health and performance of marathon runners worldwide[3] as they demand activity that places immense stress on the musculoskeletal system, increasing the risk of overuse injuries, such as stress fractures, tendinopathies, and ligament sprains. These injuries not only cause physical discomfort and impair performance but also lead to substantial downtime for athletes, affecting their training schedules, aspirations, and overall well-being [4]. Athletic runner injury prediction system is motivated by the potential impact it can create, by providing actionable insights into injury risks and warning signs, like training load management, biomechanical optimization, and injury prevention strategies and thus contribute to the promotion of athlete welfare.

B. Objective

The objective of this project is to develop an Athletic Runner Injury Prediction System using advanced machine learning techniques to improve the well-being of competitive runners by reducing the risk of injuries, which is in line with Sustainable Development Goals, ensuring healthy lives and promoting well-being.

The rest of the paper is organized as follows: Section II discusses the related works. Section III describes the methods involved in the research. Section IV presents the results, and Section V provides the conclusion of the work.

II. RELATED WORK

According to [5] a systematic review evaluating the methodological rigour of musculoskeletal injury prediction models in sports medicine is executed. The review examined thirty studies and 204 models, revealing significant methodological disparities: 60% of models relied on regression, 13% on machine learning, and 27% on both. Surprisingly, none of the models underwent external validation, raising concerns about their robustness and generalizability. Additionally, the review highlighted

shortcomings in reporting practices, such as limited details on sample size calculations, internal validation procedures, and code availability, underscoring potential biases and hindering reproducibility. Notably [6] delves into the topic of injury prediction in competitive runners using machine learning techniques. The study introduces a novel approach centred around machine learning algorithms to forecast injuries among runners, leveraging training load data as a key input. By harnessing the power of machine learning, the researchers aim to develop predictive models capable of discerning patterns and trends in training load data that may precede or indicate the likelihood of injury occurrence.

Paper [7] talks about how injuries are prevalent in sports and can lead to significant physical, psychological, and financial repercussions. Employing machine learning (ML) methods, could enhance injury prediction and facilitate effective approaches to injury prevention. Various ML techniques were utilized, including tree-based ensemble methods, Support Vector Machines, and Artificial Neural Networks. Classification methods were enhanced through preprocessing steps, over and under-sampling methods, hyperparameter tuning, feature selection, and dimensionality reduction. The research [8] says, that while sports offer numerous benefits, they also pose the risk of overuse injuries if pursued excessively, making early detection of a runner's exertion crucial for injury prevention and overall well-being. The paper [9] introduces a novel approach for predicting running fatigue by leveraging depth images, joint points, and the Borg scale. By integrating depth images and joint points, the method captures rich spatial information about the runner's movements, allowing for a comprehensive analysis of running biomechanics and posture. According to [10] managing sports performance is of utmost importance in the sports industry, with a focus on enhancing athlete performance and mitigating injury risk. Various factors, such as player health, emotional well-being, exercise load, and physical intensity requirements, contribute to these objectives. Injury prediction plays a crucial role in injury prevention, serving as a primary indicator of effective preventive measures. The proposed objective of the Artificial Neural Network (ANN) is to develop and utilize early performance and exercise load data to establish a hierarchical machine learning prediction system capable of accurately detecting player injuries.

Notably [11] a study that investigates the utilization of data from training sessions, competitions, and weather conditions for the prediction of race results and injuries in competitive running. The study implements two neural network models and evaluates them using realistic ultramarathon runner data. One model is designed to predict kilometres run for a given duration, while the other aims to classify the occurrence of injuries based on the running distance. The results demonstrate satisfactory performance for both models. The paper [12] talks about how traditional sports relied primarily on players' abilities, with limited consideration for science and knowledge. However, as sports evolved into professions and industries, the integration of technology and data analysis has become imperative for achieving objectives. In this study, classification methods for sports datasets were applied. In the prospective study [13], 355 elite male youth football players aged 10–18 underwent a pre-season neuromuscular screening protocol, including anthropometric measurements and assessments such as single leg countermovement jump (SLCMJ), single leg hop for

distance (SLHD), 75% hop distance and stick (75%Hop), Y-balance anterior reach, and tuck jump evaluation. The study compared traditional regression analysis with supervised machine learning algorithms. The paper [14] aims to leverage machine learning algorithms to predict heightened musculoskeletal injury risk post-sport-related concussions.

III. METHODS

A. Dataset

The dataset <https://doi.org/10.34894/UWU9PV> consists of a detailed training log from a prominent Dutch elite running team, covering a period of seven years from 2012 to 2019 [15]. This dataset focuses on middle and long-distance runners within the team, including athletes competing in distances from 800 meters up to the marathon.

nr. session	nr. rest days	total kms	max km o	total km Z3-Z4-Z5	nr. tough ses	i
5	2	22.2	16.4	11.8	1	
5	2	21.6	16.4	11.7	1	
5	2	21.6	16.4	11.7	1	
5	2	21.6	16.4	11.7	1	
6	1	39.2	17.6	18.9	1	
6	1	39.2	17.6	18.9	1	
6	1	33.3	17.6	15.4	1	
5	2	33.3	17.6	15.4	1	
5	2	32.9	17.6	15.5	0	
5	2	32.9	17.6	15.5	0	
5	2	32.9	17.6	15.5	0	
5	2	25.4	10.5	11.1	0	

avg recov	min recov	max recov	Athlete	IC	injury	rel total k	rel total k	rel total k	Date	
0.18	0.16	0.2	0	0	0.718447	1.378882	1.919255	0		
0.18	0.16	0.2	0	0	0.683544	1.018868	1.490566	1		
0.17	0.16	0.18	0	0	0.683544	1.018868	1.490566	2		
0.18	0.16	0.18	0	0	0.683544	1.018868	1.490566	3		
0.17	0.16	0.18	0	0	2.202247	1.361111	0.618056	4		
0.17	0.16	0.18	0	0	2.202247	1.361111	0.618056	5		
0.17	0.16	0.18	0	0	1.5	1.07767	0.718447	6		
0.17	0.16	0.18	0	0	1.5	1.07767	0.718447	7		
0.17	0.16	0.18	0	0	1.523148	1.041139	0.683544	8		
0.17	0.16	0.18	0	0	1.523148	1.041139	0.683544	9		
0.17	0.16	0.18	0	0	1.523148	1.041139	0.683544	10		
0.17	0.16	0.18	0	0	0.647959	1.426966	2.202247	11		
0.17	0.16	0.18	0	0	0.647959	1.426966	2.202247	12		
0.17	0.16	0.18	0	0	0.537538	0.806306	1.5	13		
0.17	0.15	0.18	0	0	0.537538	0.806306	1.5	14		

Fig. 1. Partial image of the dataset

Notably, the head coach overseeing the team remained consistent throughout the data collection period, ensuring continuity in training methodologies. The dataset comprises samples from a total of 74 runners, including 27 women and 47 men, providing a diverse representation of both genders within the high-level running context. The dataset as shown in Fig. 1 includes the following columns.

For the weekly overview, the summarised features include:

- Total sessions: The overall number of training sessions completed.
- Rest days: The count of days without any training.
- Total distance: The cumulative running mileage covered.
- Max distance in a day: The longest distance covered in a single day.
- Total km in Z3-Z4-Z5-T1-T2: The distance covered at or above the aerobic threshold, including speeds in Z3 or faster.
- Tough sessions: The number of high-intensity sessions in Z5, T1, and T2, indicating efforts above the anaerobic threshold or intensive track intervals.
- Interval session days: The days featuring a Z3 or faster training session.

- Total km in Z3-4: The distance covered in the Z3- Z4 range, which falls between the aerobic and anaerobic thresholds.
- Max Z3-4 distance in a day: The farthest distance covered in the Z3-Z4 range within a single day.
- Total km in Z5-T1-T2: The total distance covered at Z5, T1, and T2 speeds.
- Max Z5-T1-T2 distance in a day: The longest distance covered at Z5-T1-T2 speeds in a single day.
- Cross-training hours: The total time spent on alternative training methods.
- Strength training sessions: The total number of strength training sessions completed.
- Average exertion: The mean rating of perceived exertion for all training sessions.
- Minimum exertion: The lowest perceived exertion rating across all sessions for the week.
- Maximum exertion: The highest perceived exertion rating across all sessions for the week.
- Average training success: The mean rating of how successful each training session felt to the athlete.
- Minimum training success: The lowest training success rating for the week.
- Maximum training success: The highest training success rating for the week.
- Average recovery: The mean rating of how well-rested the athlete felt before each session.
- Minimum recovery: The lowest restfulness rating before any session during the week.
- Maximum recovery: The highest restfulness rating before any session during the week.

B. Proposed Method

The algorithms Support Vector Machine (SVM)[16] [17], Bagging, XGBoost Decision Tree, Multi-Layer Perceptron[18], Recurrent neural networks [19] and Long-short term Memory [20] were used to predict injuries. The architecture of the proposed model is shown in Fig.2. The dataset used comprised 42,799 data points, with 575 instances of injury and 42224 representing non-injury cases. To address the class imbalance, randomly select a subset of instances from the majority class (non-injury) to match the number of instances in the minority class (injury). By ensuring a more equitable representation of both classes in the training data, the research aimed to prevent the model from being biased towards the majority class and improve its ability to learn from the minority class instances. To further address the imbalance in the dataset, the research augmented the minority class (injury) by adding 575 instances, resulting in a balanced dataset comprising 1150 instances of both injury and non-injury classes. This augmentation ensured an equal representation of both classes

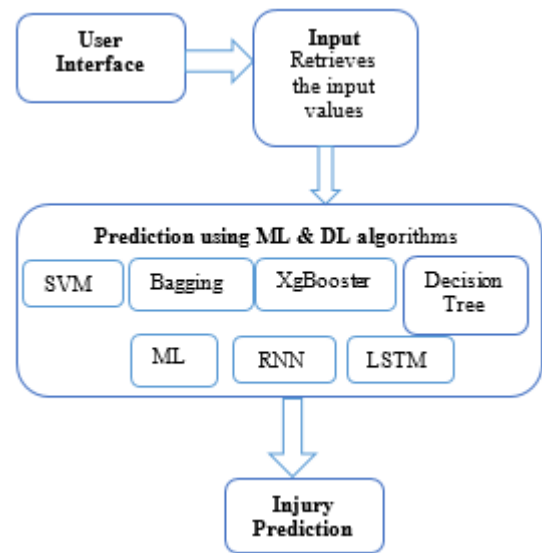


Fig. 2. Architecture of the Proposed System

In the training data, facilitating more robust model learning without the need for down sampling the majority class. To further improve class balance, the research augmented the minority class (injury) by adding 575 instances, resulting in a balanced dataset comprising 1725 instances of both injury and non-injury classes. This augmentation ensured a more equitable representation of both classes in the training data, mitigating potential biases and enhancing the model's ability to learn meaningful patterns from both classes. To achieve greater class balance, the minority class (injury) was augmented by incorporating an additional 575 instances. Consequently, the dataset now comprises 2300 instances, evenly distributed between injury and non-injury classes.

The minority class (injury) underwent augmentation by adding 575 instances. As a result, the dataset expanded to encompass 2875 instances, with an equal distribution between injury and non-injury classes. This augmentation facilitated a more balanced representation of both classes during model training, effectively mitigating potential biases and enhancing the model's ability to discern meaningful patterns from both categories. A standardized procedure was followed for building the model. The balanced dataset was divided into training and validation subsets, and the algorithm was trained on the training subset. The trained model's performance was then evaluated on the validation set, calculating metrics such as accuracy, precision, and F1 Score to assess its effectiveness in injury classification. Throughout this process, an iterative analysis of the model's performance was conducted, adjusting parameters or employing feature engineering techniques as necessary to improve performance. For ease of user, the work is developed with GUI where the user inputs all the metrics of the runner. Once the inputs are entered, the system will analyse the prediction of injury and give the result as shown in Fig.14.

Visualisations such as histograms, scatter plots, and heatmaps provided valuable insights into the distribution of features, potential correlations, and any underlying patterns that could inform the modelling efforts

IV. RESULTS

With a clear understanding of the dataset, appropriate evaluation metrics were selected to assess the performance of the predictive models comprehensively. The evaluation metrics such as confusion matrix, ROC curve, accuracy, Precision and F1 Score were used to evaluate the models' predictive capabilities from different perspectives. The confusion matrix provided insights into the models' classification performance, while the ROC curve helped assess their ability to trade off true positive rate and false positive rate. Accuracy served as a general measure of model performance, providing an overall assessment of predictive accuracy. The outcomes of this comprehensive evaluation provide invaluable insights into the performance of diverse machine learning and deep learning algorithms on an augmented dataset of injury and non-injury classes. The confusion matrix depicted in Fig.3,4,5,6 shows the algorithm's adeptness in classifying a substantial number of instances as true positives and true negatives, indicative of its capacity to effectively discern both injury and non-injury cases. To execute this project, PyCharm was to installed which requires a minimum of 8 GB RAM.

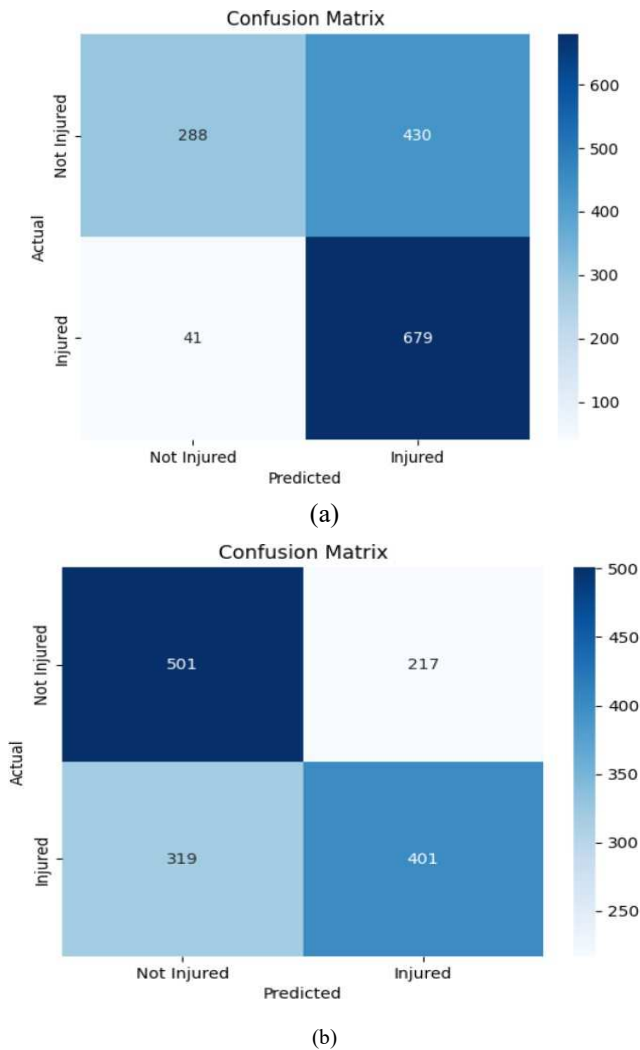


Fig. 3. Confusion matrix for SVM, Confusion matrix for SVM (a), Bagging (b)

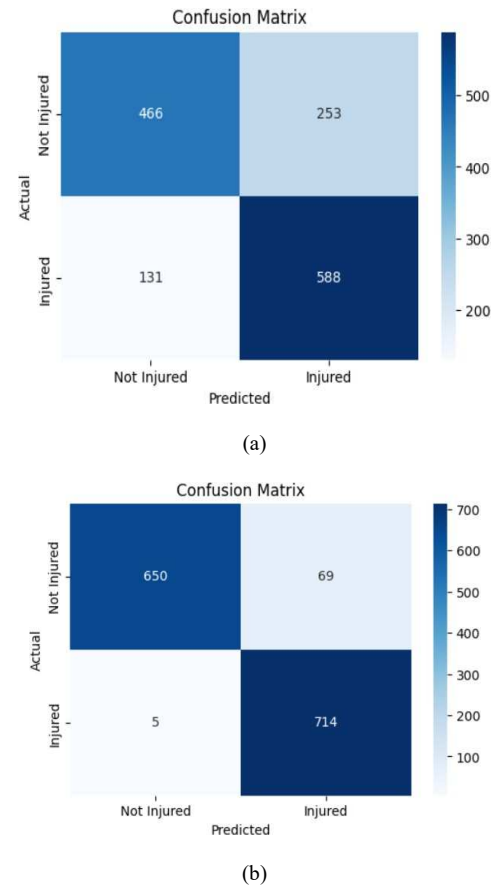


Fig. 4. (a) Confusion matrix for XGBoost, (b) Confusion matrix for Decision Tree

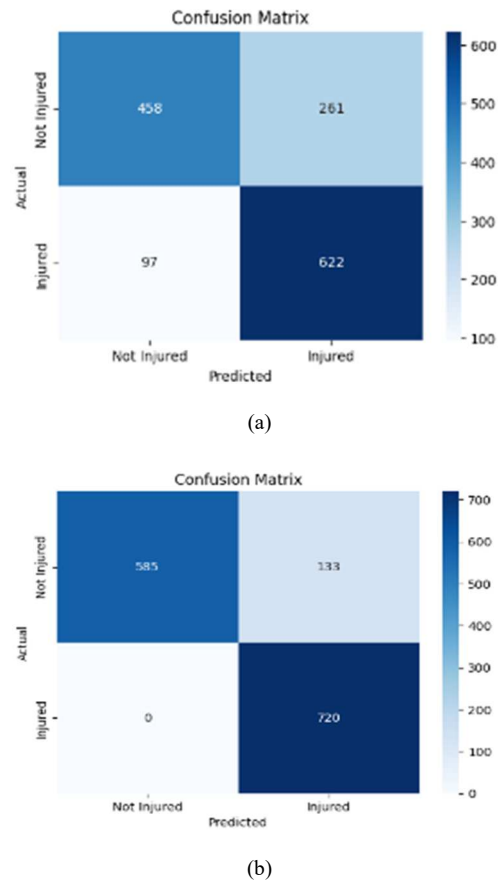


Fig. 5. (a) Confusion matrix for MLP, (b) Confusion matrix for RNN

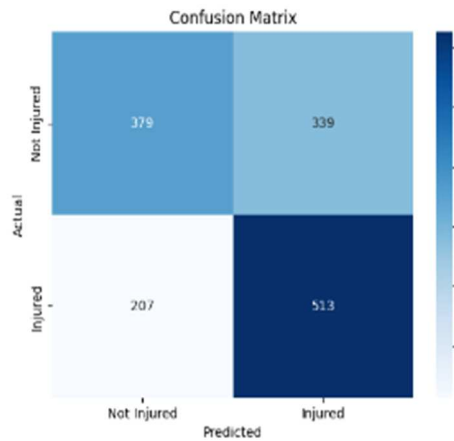


Fig. 6. Confusion matrix for LSTM

The validation curve for the SVM Classifier is shown in Fig. 7, In Fig. 8, the validation curve for the Bagging Classifier is depicted, the training score is near to 1.00 and the cross-validation score is 0.93. Fig. 9 shows the Validation curve for the XgBoost Classifier. Fig. 10 visualizes the validation curve for the decision tree classifier. Here the training and cross-validation curves are very close. Fig. 11 shows the validation curve for the MLP classifier. The training and cross-validation scores are 0.64 and 0.65 respectively. In Fig. 12 Validation curve for the RNN classifier is plotted. The validation curve for the LSTM classifier is shown in Fig. 13.

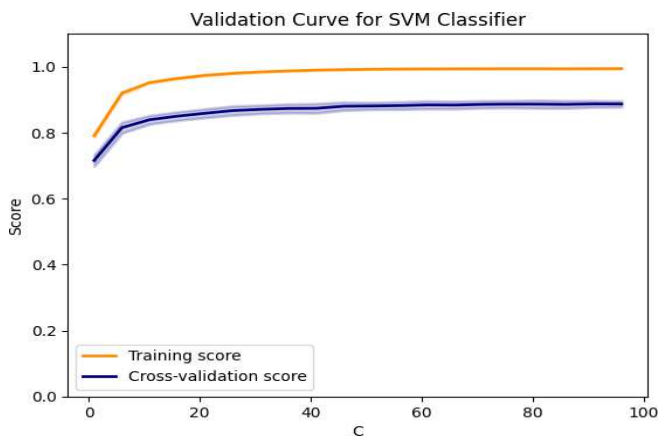


Fig. 7. Validation curve for SVM Classifier

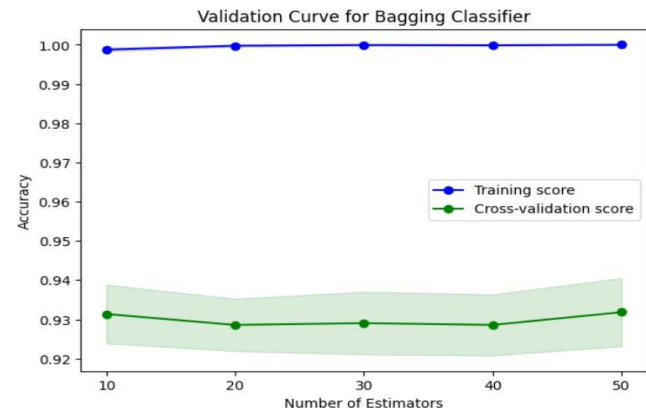


Fig. 8. Validation curve for Bagging Classifier

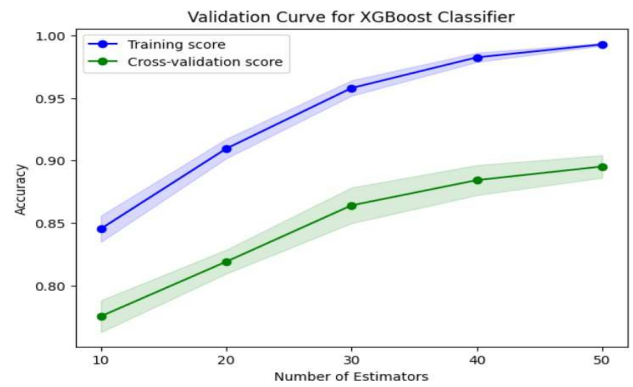


Fig. 9. Validation curve for XGBoost Classifier

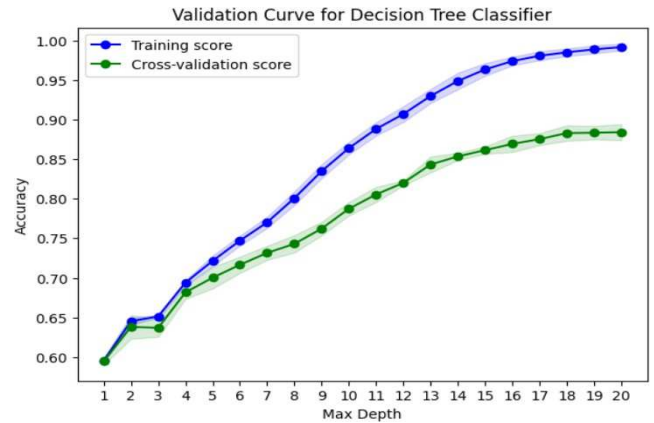


Fig. 10. Validation curve for Decision Tree Classifier

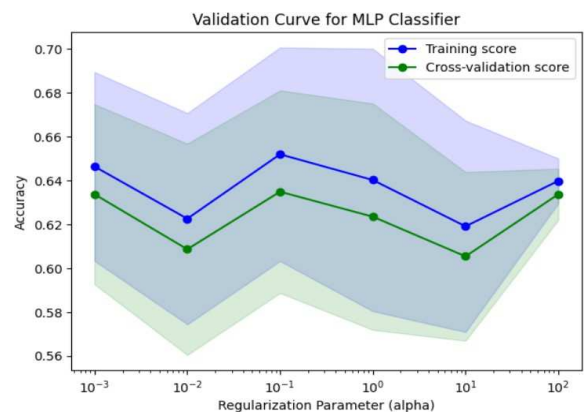


Fig. 11. Validation curve for MLP Classifier

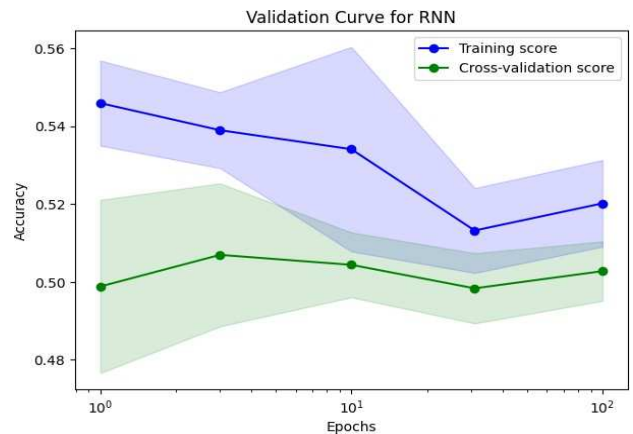


Fig. 12. Validation curve for RNN Classifier

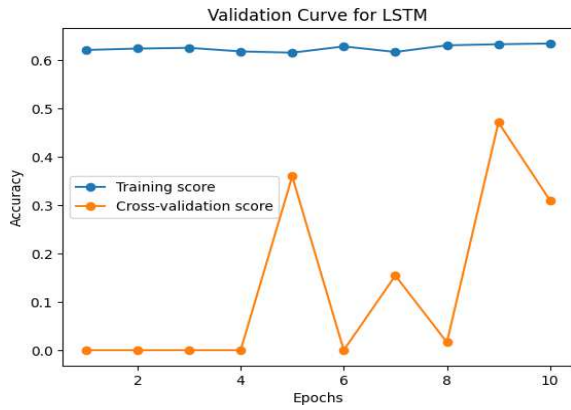


Fig. 13. Validation curve for LSTM Classifier

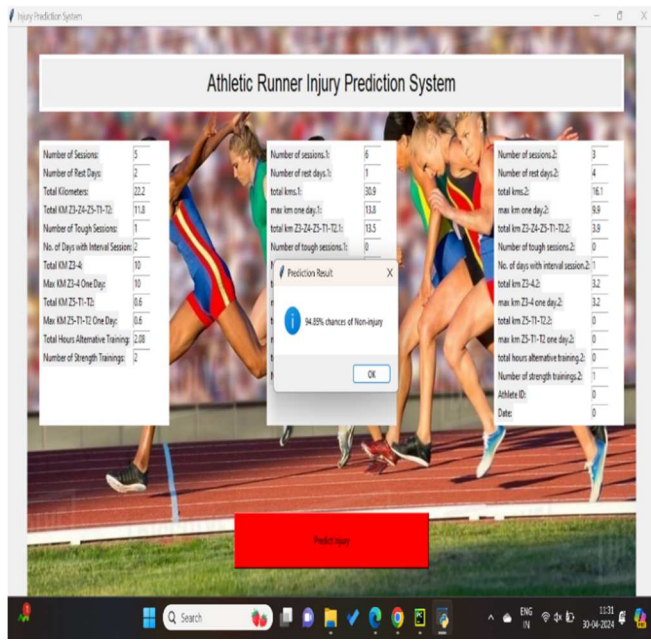


Fig. 14. Application's GUI

TABLE I. PERFORMANCE METRICS -SVM,BAGGING

Sample Size	SVM			Bagging		
	Accuracy	Precision	F1 Score	Accuracy	Precision	F1 Score
575*4224	0.94	0.03	0.05	0.98	0	0
575*575	0.58	0.57	0.6	0.57	0.57	0.6
1150*1150	0.65	0.63	0.67	0.78	0.74	0.8
1725*1725	0.7	0.68	0.71	0.88	0.83	0.89
2300*2300	0.7	0.66	0.73	0.92	0.88	0.92
2875*2875	0.73	0.69	0.75	0.94	0.91	0.95

TABLE II. PERFORMANCE METRICS -XGBOOSTER,DT

Sample Size	XGBooster			Decision Tree		
	Accuracy	Precision	F1 Score	Accuracy	Precision	F1 Score
575*4224	0.98	1	0.01	0.97	0.03	0.04
575*575	0.68	0.65	0.71	0.58	0.53	0.58
1150*1150	0.71	0.67	0.74	0.79	0.76	0.81
1725*1725	0.72	0.68	0.75	0.85	0.8	0.86
2300*2300	0.73	0.69	0.75	0.89	0.83	0.9
2875*2875	0.71	0.68	0.74	0.92	0.86	0.92

TABLE III. PERFORMANCE METRICS- MLP, RNN

Sample Size	MLP			RNN		
	Accuracy	Precision	F1 Score	Accuracy	Precision	F1 Score
575*4224	0.87	0.02	0.04	0.94	0.01	0.03
575*575	0.58	0.57	0.45	0.58	0.55	0.55
1150*1150	0.51	0.59	0.3	0.65	0.6	0.73
1725*1725	0.63	0.62	0.66	0.6	0.56	0.72
2300*2300	0.54	0.73	0.28	0.57	0.66	0.42
2875*2875	0.68	0.62	0.74	0.6	0.62	0.57

TABLE IV. PERFORMANCE METRICS – LSTM

Sample Size	LSTM		
	Accuracy	Precision	F1 Score
575*4224	0.95	0.03	0.05
575*575	0.5	0.4	0.6
1150*1150	0.64	0.62	0.7
1725*1725	0.62	0.59	0.71
2300*2300	0.61	0.6	0.65
2875*2875	0.61	0.58	0.69

V. DISCUSSIONS

The results in Table I, II and III,IV suggest that Bagging and Decision Tree are promising algorithms for predicting athletic runner injuries, with potentially being more conservative in their predictions. However, further analysis and consideration of other factors, such as computational efficiency and interpretability, are warranted for comprehensive evaluation and selection of the most suitable algorithm. The performance of the algorithms varied across different sizes in the dataset. Notably, models trained using Bagging and Decision Tree consistently showed the defined thresholds for accuracy and precision, indicating their robustness in predicting athletic runner injuries. The analysis of the results reveals a notable trend where machine learning (ML) algorithms demonstrated comparatively better performance than deep learning (DL) algorithms across various performance metrics. This observation can be attributed to the nature of the dataset, which predominantly comprises numerical features.

VI. CONCLUSION

The Athletic Runner Injury Prediction System has addressed a critical need in sports science by developing a cutting-edge tool for predicting and preventing injuries in athletic runners. By leveraging comprehensive datasets, incorporating biomechanical insights, and employing

machine learning algorithms, the system has demonstrated its ability to forecast potential injuries. The user-friendly interface ensures practicality for athletes and their support teams, facilitating informed decision-making in training protocols. The future scope for the Athlete Runner Injury Prediction System holds immense potential for transformative advancements in sports science. Expanding beyond current capabilities, the system could incorporate a multimodal approach, integrating diverse data sources for a more comprehensive understanding of an athlete's health. Collaboration with sports medicine professionals and continuous refinement of machine learning models would enhance the system's accuracy and applicability.

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